Development of a MATLAB-Based Vapor-Liquid Equilibrium and Flash Separation Simulator for an Ethanol-Water System

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Abstract

This project designed a MATLAB simulator for vapor-liquid equilibrium, called VLE throughout the report, and flash separation of an ethanol-water mixture. Ideal liquid solution was assumed to use Raoult's law to describe phase equilibrium, and a certain temperature range was assumed to utilize Antoine's equation to find the component saturation pressures. To characterize the phase behavior of the mixture, bubble point and dew point were calculated and graphed, and the K-values and Rachford-Rice equation determined phase condition and vapor fraction in flash separation. The MATLAB simulation then investigated the individual effects of different operating conditions (temperature, pressure, feed composition) on phase behavior and outlet streams. The results found that the operating conditions heavily influenced the liquid-vapor split, with a higher temperature favoring the vapor phase and a higher pressure favoring the liquid phase. Additionally, the project focused on building useful MATLAB functions and numerical methods for chemical engineering computations. Though the simulator is a helpful representation of an idealized flash separation, it has a lot of limitations. It assumes ideal liquid phase behavior and a specific range of Temperature to satisfy the Antoine equation boundaries. Future projects or improvements could take into account nonideal activity coefficient models, energy balances, multicomponent mixtures, and more. To sum it all, the project is intended to demonstrate how thermodynamic principles can be applied computationally for analyzing and predicting a separation process.

1.1 Background

Vapor-liquid equilibrium (VLE) is the state in which the condensation rate is equal to the evaporation rate, equaling no net transfer of material between phases. Therefore, VLE is a dynamic equilibrium, not a state in which molecular movement has stopped.

VLE occurs in pure substances simpler than mixtures. For pure substances, it occurs when the liquid's saturation vapor pressure is equal to the total system pressure. The boiling point is fixed for a given pressure and the composition of both phases stays the same. VLE behavior is more complex for mixtures as the components of a mixture can have different volatilities. This results in the compositions of the liquid and vapor phases being different at equilibrium, and the difference becomes the groundwork for separation processes like flash separation and distillation.

VLE is applied through flash separation in this project. Flash separation is a single stage separation process that partially separates a mixture based on the differences of different components' volatility. This principle is used in various processes in the oil and gas industry, like petroleum refining to separate hydrocarbon streams, or natural gas processing to split between condensed hydrocarbons from the gas stream.

Simulating VLE and flash systems is important to predict whether the process stream will be a vapor, liquid, or both, and to figure out the compositions and amounts of each phase. These are important to know in order for engineers to model the effects of temperature, pressure, and feed composition and predict the process behavior before changing anything in the system.

1.2 Project Objectives

The main objective of this project was to develop a MATLAB program to simulate the VLE and flash separation of a mixture consisting of ethanol and water while applying, and building upon, fundamental thermodynamics knowledge.

The program was designed to accept reasonable operating conditions, like: temperature, pressure, feed flow rate, and feed composition to figure out the phase behavior of a mixture. The program finds whether the mixture exists as a vapor, liquid, or two-phase system at the said conditions by calculating the saturation pressures of ethanol and water and computing the equilibrium ratios. When it is possible, the program calculates the vapor fraction, flow rates, and compositions of the liquid and vapor streams.

1.3 Scope and Assumptions

The project's scope focused on the VLE and flash separation of a binary ethanol and water mixture. As for the program, it was made to discover the phase behavior of the mixture at very specific conditions, sometimes inputted by the user, and calculate the final liquid and vapor streams. The program inputs included pressure, temperature, feed flow rate, and ethanol/water feed composition. Additionally, sensitivity studies were held to see the effect of different pressure, temperature, and feed composition on outlet flow rates and vapor fraction. It was a binary system, so it was not intended to simulate multicomponent mixtures.

Many assumptions were made for this project. The liquid mixture was taken into account as an ideal solution and Raoult's law was used to model it, and the vapor phase was assumed to act ideally too. For vapor pressures, they were assumed using the Antoine equation, which relates saturation vapor to temperature. Since the Antoine equation is valid within a specific temperature range for its constants, the accuracy of the calculated saturation pressures depends on said defined limits. Finally, the model also does an isothermal flash calculation. This is done by specifying the pressure and temperature rather than determining them through an energy balance, hence heat transfer and enthalpy changes not being counted.

2.1 Vapor-Liquid Equilibrium [1] [2]

Vapor-Liquid Equilibrium (VLE) lays the groundwork for the rest of the project. It is the state in which the condensation rate equals the evaporation rate, meaning liquid and vapor coexist and there is no material net transfer between the phases, representing a dynamic equilibrium. The molecules in the liquid constantly gain enough energy to enter the vapor phase through evaporation, while the molecules in the vapor hit the liquid surface and go to the liquid phase by condensation.

Given a mixture contains various components, every component also does this equilibrium and distributes itself between the vapor and liquid phase: the components with the greater volatility favor the vapor phase, while the components with the lower volatility favor the liquid phase. The difference in compositions of liquid and vapor phases allows VLE to be used for separation processes like the flash separation in this project.

2.2 Antoine Equation [3] [4] [5]

The Antoine Equation, which was used throughout nearly the entirety of the project, is used to find the saturation vapor pressure of a pure substance as a function of temperature. It relates vapor pressure to temperature and is an empirical correlation made up from experimental vapor pressure data. The equation is: $log_{10}(P_i^{sat}) = A_i - B_i/(C_i + T)$ where P_i^{sat} is the saturation vapor pressure of component i, T is temperature, and A_i, B_i, and C_i are experimentally observed Antoine constants specific to each component, in this project's case for water and ethanol. Antoine constants are valid over a specific range of temperatures, hence this project having a few select temperatures.

The saturation vapor pressure is the pressure released by a vapor when in equilibrium with the liquid phase at a certain temperature. It shows a substance's tendency to vaporize: the higher the saturation vapor pressure, the higher tendency the substance has to enter the vapor phase. Knowing the saturation vapor pressure is important for this project as it is used to find how every component distributes between either phase at specific pressures and temperatures. The saturation vapor pressures are then used for various calculations in the project, these being Raoult's law, bubble point, dew point, equilibrium ratios, and flash separation calculations.

2.3 Raoult's law [1] [2]

Raoult's law is represented as $P_i = x_i P_i^{sat}$, where P_i is the partial pressure of component i, x_i is the mole fraction of that component in the liquid phase, and P_i^{sat} is its saturation vapor pressure at the system temperature in an ideal liquid mixture at VLE. It is used to find the partial vapor pressure of each component. Partial pressure is a share of the total pressure of the vapor mixture. The total vapor pressure is found by summing the partial pressures: $\sum P_i$. Figuring out total vapor pressure made by the liquid mixture is essential to the project as it can be compared to the specified system pressure to determine the phase behavior of the mixture. That information can then be used to find the bubble point.

Raoult's law shows that the partial pressure is dependent on the concentration in the liquid as well as its tendency to vaporize at specific temperatures. Raoult's law can be combined with the total system pressure to relate the vapor and liquid compositions of each component when in equilibrium. Raoult's law assumes ideal behavior in the mixture, simplifying the VLE relationship in the project.

2.4 Bubble Point [1] [2]

The bubble point is the condition at which the first amount of vapor forms from a liquid mixture. When a mixture is below its bubble point, at a specific pressure, it is a liquid. Heating the mixture increases the saturation vapor pressures of the components in the mixture. The first bubbles of vapor that form are a sign of when the bubble point is reached, and that happens when the combined partial pressures of the components become equal to the system pressure.

For an ideal mixture following Raoult's law, the bubble point condition at a specific pressure is $P = \sum_i x_i P_i^{sat}(T)$ where P is the total system pressure, x_i is the liquid-phase mole fraction of component i, and $P_i^{sat}(T)$ is its saturation vapor pressure of component i at temperature T.

For a binary ethanol-water mixture, the equation becomes $P = x_E P_E^{sat}(T) + x_W P_W^{sat}(T)$, where the component E is ethanol and the component W is water.

A greater portion of the liquid mixture vaporizes as heating continues after the bubble point and within the two phase region.

2.5 Dew Point [1] [2]

The dew point is the opposite of the bubble point, in the sense that the dew point is the condition at which the first amount of liquid forms from a vapor mixture. When a mixture is above its dew point, at a specific pressure, it is in the vapor phase. Cooling the mixture decreases the saturation vapor pressure of the mixture’s components, forming the first drops of liquid.

For an ideal mixture, the dew point condition at a specific pressure can be written as $\sum_i \frac{y_i P}{P_i^{sat}(T)} = 1$ where y_i is the vapor-phase mole fraction of component i, P is the total system pressure, and $P_i^{sat}(T)$ is the saturation vapor pressure of the component i at temperature T.

For the binary ethanol-water system, $\frac{y_E P}{P_E^{sat}(T)} + \frac{y_W P}{P_W^{sat}(T)} = 1$, where component E is ethanol and component W is water. Cooling the mixture below the dew point increases the portion of the vapor that condenses. Given that the dew point and the bubble point are opposites of each other, they are used as the boundaries of the two-phase region to signify where vapor and liquid coexist.

2.6 K-Values [1] [2]

The K-Value refers to the equilibrium ratio. It shows how much of the component is concentrated in the vapor phase to the liquid phase. Generally speaking, it is important as it aid in predicting whether a mixture will mostly boil or stay in the liquid phase.

It is defined as $K_i = \frac{P_i^{sat}}{P}$ where y_i and x_i are the liquid and vapor phase mole fractions of component i. As per the assumptions of Raoult’s law for an ideal mixture, the K-value can be calculated as $K_i = \frac{P_i^{sat}}{P}$ where P_i^{sat} is the component's saturation vapor pressure and P is the total system pressure.

The result of the K-Value shows which phase the component favors. When K>1, the component favors the vapor phase, and when K<1, the component favors the liquid phase. When K=1, it means that the component has the same mole fraction in both vapor and liquid phases at equilibrium.

2.7 Flash Separation [1] [2]

Flash separation is the process that splits the mixture, called the feed, into vapor and liquid streams. In this project's case, the feed was a mixture of water and ethanol. Prior to entering the process, the feed is brought to a specific temperature and pressure then is put into the vessel: the flash separator. If part of the mixture exists as a vapor while the remaining exists as a liquid, it means the conditions fall within the two phase region.

The flash separator allows the liquid and vapor phases to separate through gravity and the components' density. Components more likely to vaporize are more concentrated in the vapor phase, and the components more likely to remain liquid stay in the liquid phase, and each resulting fluid goes through its own outlet. Ethanol is generally more volatile, meaning it has a higher tendency to vaporize.

The amount of vapor/liquid produced from the process is dependent on the feed's composition (ethanol/water), temperature and pressure, and the equilibrium behavior of the components present. A flash separator is a single stage equilibrium process, meaning it is not like a distillation column, as a distillation column uses repeated equilibrium stages for separation.

2.8 Rachford-Rice Equation [2] [6]

The Rachford-Rice equation is an equation used to calculate the vapor fraction, denoted as β (beta), which can find phase compositions. The vapor fraction is essential to help find how a feed divides the vapor and liquid phases during flash separation. The vapor fraction is the section of the total feed that exits the flash process as vapor and is defined as $\beta = \frac{V}{F}$ where V is the vapor flow rate and F is the total feed flow rate. The remaining fraction, $1 - beta$, represents the portion of the feed that exits as liquid.

As for a feed with multiple components, material balance and individual component balances must be satisfied. In addition, the vapor and liquid components must satisfy VLE. Combining all the relationships with the equilibrium ratios provides the Rachford-Rice equation:

$\sum_i \frac{z_i(K_i - 1)}{1+\beta(K_i - 1)} = 0$, where z_i is the overall feed mole fraction of component i, K_i is its equilibrium ratio, and β is the vapor fraction. The equation can be solved numerically for β given the feed composition and K-values are known at a specific temperature and pressure.

A two phase flash requires the vapor fraction to be between zero and one. When the value is closer to zero, it means more of the feed exists as a liquid. When it is closer to one, more of the feed exists as a vapor. The Rachford-Rice equation can be evaluated at a boundary between $\beta = 0$ and $\beta = 1$. The left side of the Rachford-Rice equation can be thought of as f(β). With that in mind, if f(0) and f(1) do not have the same signs, it means a physical two phase flash is possible and that the physical range has a root. In the program, a test was done to find whether the two phase flash was possible at the given conditions. The specific temperature, pressure, and feed composition conform to a single phase condition rather than a two phase condition when there is no root.

```
if f0 > 0 && f1 < 0

    % Two-phase flash
    phase = "Two Phase";
    beta_guess = 0.50;
    options = optimoptions('fsolve','Display','off');
    beta_val = fsolve(f,beta_guess,options);

    V = beta_val * F;
    L = (1 - beta_val) * F;
    x_ethanol = z_ethanol / (1 + beta_val * (K_ethanol - 1));
    x_water = z_water / (1 + beta_val * (K_water - 1));

    y_ethanol = K_ethanol * x_ethanol;
    y_water = K_water * x_water;
elseif f0 <= 0 && f1 <= 0

    % All liquid
    phase = "All Liquid";
    beta_val = 0;
    V = 0;
    L = F;
    x_ethanol = z_ethanol;
    x_water = z_water;

elseif f0 >= 0 && f1 >= 0

    % All vapor
    phase = "All Vapor";
    beta_val = 1;
    V = F;
    L = 0;
    y_ethanol = z_ethanol;
    y_water = z_water;

end
```

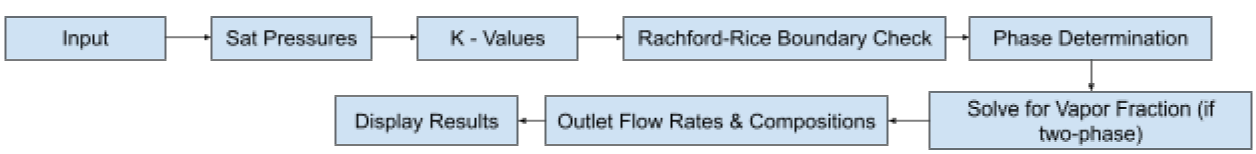
Once β has been computed, the liquid phase composition of each component can be calculated with the following equation: $x_i = \frac{z_i}{1+\beta(K_i - 1)}$, and the vapor phase composition can be calculated from $y_i = K_i x_i$. The vapor and liquid flow rates are calculated as $V = \beta F$ and

$L = (1 - \beta)F$ where L is the liquid flow rate.

3.1 Overall Program Structure [2]

The MATLAB program takes user inputted data, like the feed composition, feed flow rate, system temperature and pressure. Said data is passed through a series of different calculations to carry out VLE and flash separation calculations mentioned in the previous sections. The calculations determine the phase behavior of the mixture and the resulting vapor and water streams if a two phase flash is possible.

The program starts off by calculating the saturation vapor pressures of ethanol and water at a specified temperature with the Antoine equation. The saturation pressures and specified system pressure are used to calculate the K-values of each component, which are then used in the Rachford-Rice function, along with the feed composition, to determine if a physical two phase solution exists. The program solves for the vapor fraction and the result is used to calculate the vapor and liquid flow rates and their respective compositions if a two-phase flash is possible. The calculated results are then displayed for the user. A visual representation of the structure of the program is below.



3.2 MATLAB Functions

Two MATLAB functions were made in this program: *antoinePressure.m* and *flashSeparator.m*. Starting off with antoinePressure.m, it was used to calculate the saturation vapor pressure of a component with the Antoine equation. Its parameters are *T, A, B* and *C*, where *T* is the temperature in Celsius, and A, B, and C being the three Antoine constants. The same function can be given to both water and ethanol separately without rewriting the equation given they require different constants.

As for *flashSeparator.m*, it is where the entire flash separation calculations were stored. It came towards the end of the program. This function takes the required information regarding the feed and operating conditions and computes the phase behavior and outlet conditions. This singular function follows the program structure mentioned in section 3.1, making individual calculations easier to separate into reusable parts.

3.3 Flash Separator Algorithm [2] [6]

The flash separator algorithm begins with the operating conditions: temperature, pressure, feed flow rate, and ethanol feed composition. As for water feed composition, it can be found by finding the remaining mole fraction to total up to 1. Then the saturation vapor pressures for ethanol and water at the inputted temperatures are found by calling *antoinePressure.m*, a script in the project folder containing the antoine equation.

```
function P = antoinePressure(T,A,B,C)
P = 10.^(A - (B ./ (T + C)));
end
```

The saturation pressures are important to know as they are then used to find the K-value by dividing the saturation pressures by the inputted system pressure. The K-values are used alongside the feed compositions to compute the Rachford-Rice equation at $\beta = 0$ and $\beta = 1$. This check is used to find if a physical two phase root exists between the 0 and 1 boundaries: all vapor and all liquid boundaries. If the function changes sign between 0 and 1 it means a two phase flash is possible and the Rachford-Rice equation outputs the physical vapor fraction.

The physical vapor fraction is then stored as β. After this step, the liquid phase mole is calculated using β, the K-value, and the feed compositions. Then, the vapor phase mole fractions are computed using the equilibrium relationship between the liquid and vapor compositions. A check is executed by summing every phase composition to equal 1.

Nearing the end of the program, the vapor and liquid flow rates are calculated. This can be found with the flow rate and vapor fraction. For the vapor flow rate $V = \beta F$, and for the liquid flow rate $L = (1 - \beta)F$. The formulae mentioned output the phase split and their compositions, allowing for evaluations to be done on the specific operating conditions for the flash separation.

3.4 Sensitivity Studies [1] [2]

Sensitivity studies were carried across the different operating conditions to determine how their changes affect the phase behavior and flash separation of the mixture. The operating conditions studied were pressure, temperature, and feed composition. Each variable changes the behavior of the system including the vapor fraction, vapor and liquid compositions, and finding if a two phase flash is possible.

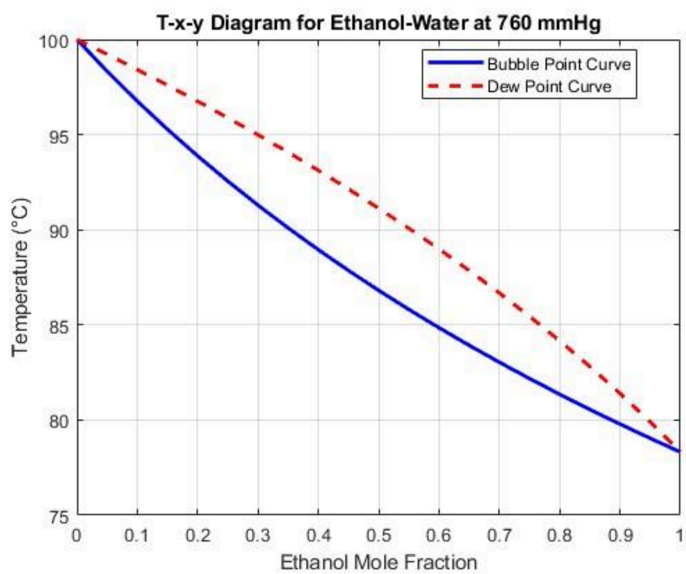
For temperature, it was varied because saturation vapor pressure increases as temperature increases. Altering the temperature directly influences the tendency of ethanol and water to enter the vapor phase, given the K-values are computed from saturation vapor pressure and system pressure. The study was helpful to find when the feed stays as a liquid, becomes vapor, enters the two phase region, or finds out how the vapor fraction changes as the mixture's temperature changes as well.

For pressure, it was also studied given it can affect the K-value as well as the components' tendencies to vaporize. Increasing the system pressure at a fixed temperature means the K-value will decrease, favoring the mixture turning to a liquid. This is important as it shows how flash separation can be influenced by changing the operation pressure.

Finally, for the feed composition, it was studied to see how the amounts of water and ethanol entering the flash separator could alter the outlet compositions. Ethanol and water have different volatilities, meaning one is more likely to change phases than the other. Ethanol being more volatile in the project's conditions, changing the ethanol mole fraction in the feed changes how much material enters the vapor phase, and how the vapor and liquid outlet compositions vary.

4.1 Bubble Point Curve

Early into the program, the bubble point curve was graphed. The calculations were to find the temperature where an ethanol-water mixture starts to form vapor at 760 mmHg. The bubble point temperatures were plotted against the ethanol mole fraction to show how vaporization changes alongside the composition.



The graph proves that the bubble point temperature changes along the composition of the liquid mixture. The bubble point temperature decreases as the ethanol mole fraction increases because ethanol is more volatile than water. Hence, a liquid mixture containing a higher amount of ethanol would have a lower boiling point than one consisting mainly of water. Ethanol contributes a larger amount of the total vapor pressure as the amount of ethanol in the liquid mixture increases.

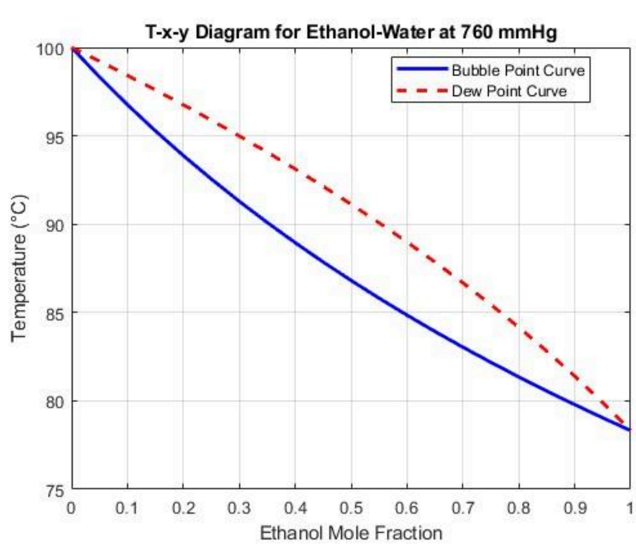
This behavior is consistent with the Raoult's law relationship:

$P = x_E P_E^{sat}(T) + x_W P_W^{sat}(T)$, where x_i is the mole fraction of component i in the liquid phase, P_i^{sat} is the saturation vapor pressure of component i, and T is temperature. The bubble point signifies when the total vapor pressure computed from the liquid composition and the component saturation pressure is equal to the system pressure. Increasing x_E means there will be a greater amount of ethanol to the total vapor pressure, ultimately reaching the specified pressure at a lower temperature.

The curve also shows the effect of the composition on the behavior of phases. For instance, the mixture with a higher amount of water needs to be heated to a higher temperature to reach vaporization, while the one with a higher content of ethanol vaporizes at a lower temperature.

4.2 Dew Point Curve

The dew point calculations were used to find the temperature where the first amount of liquid forms from an ethanol-water vapor mixture at a system pressure of 760 mmHg, the complete opposite of bubble point. The dew point temperatures were plotted against the vapor phase ethanol mole fraction.



The results show that the dew point temperature changes alongside the vapor phase. The dew point temperature decreases as the ethanol mole fraction in the vapor increases given that ethanol is more volatile than water. Therefore, a vapor mixture with a larger amount of ethanol can stay in the vapor phase at a lower temperature before the start of condensation.

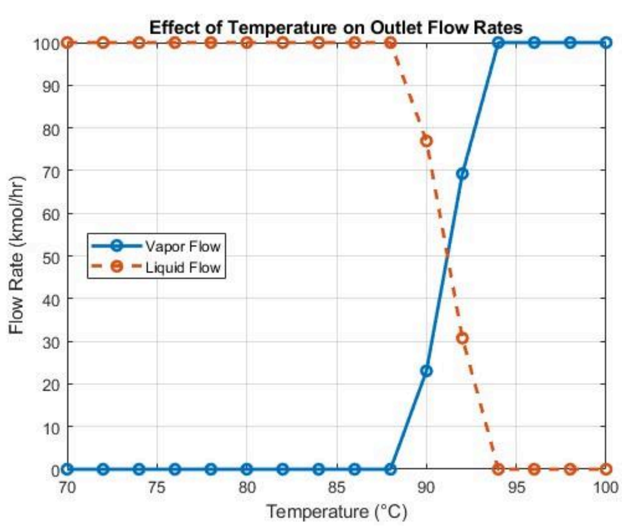
The dew point was calculated using the relationship $\sum\left(\frac{y_i P}{P_i^{sat}(T)}\right) = 1$, and for the ethanol-water mixture it is represented as $\frac{y_E P}{P_E^{sat}(T)} + \frac{y_W P}{P_W^{sat}(T)} = 1$, where y is the mole fraction of component i in the vapor phase, P_i^{sat} is the saturation vapor pressure of component i, and T is the temperature.

The dew point signifies the vapor mixture's beginning of condensation. The more it cools, the more liquid forms, the more the system moves towards a two phase region.

Together, the bubble point and dew point curves define the phase behavior of the ethanol-water mixture. The bubble point shows the point at which vaporization begins, and the dew point shows the point at which condensation begins. The region between these two curves represents the conditions where liquid and vapor can coexist.

4.3 Temperature Study

The effect of temperature on the flash separation was found by using varying values of temperature for the system while keeping the pressure (760 mmHg), feed composition (0.40 ethanol), and feed flow rate (100 *kmol/hr*) constant. The temperature study was used to find how temperature affects the equilibrium ratios, phase behavior, vapor fraction, and outlet stream flow rates.



The saturation vapor pressures of ethanol and water increase as temperature increases. Since the equilibrium ratio is calculated using

$K_i = \frac{P_i^{sat}}{P}$, an increasing saturation vapor pressure gives a larger K-value at constant system pressure. Larger K-values are signs that the components have a higher tendency to enter the vapor phase.

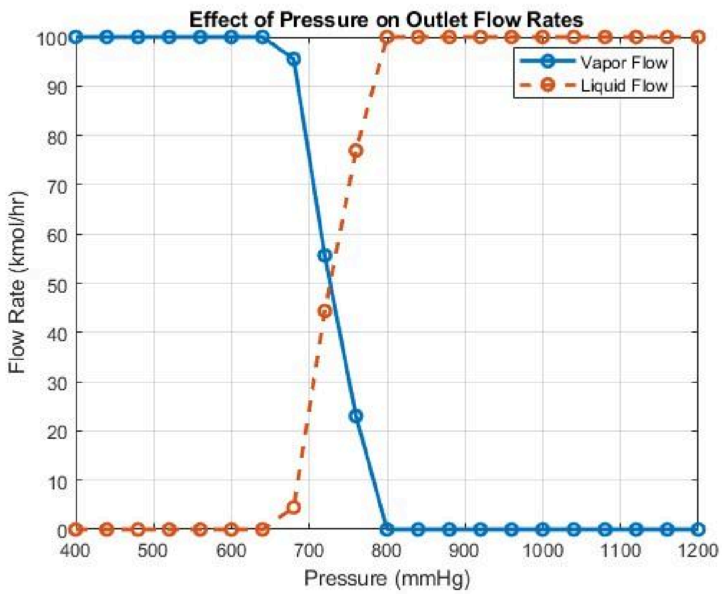
The Rachford–Rice boundary calculations demonstrate this transition well. At 85°C, the calculated values were approximately $f(0) = -0.1389$ and $f(1) = -0.3615$. Since both values have the same sign, the Rachford–Rice function does not cross zero in the physical boundary $\beta = 0$ and $\beta = 1$. Therefore, a two phase solution does not exist at the given conditions.

However, at 90°C, the calculated values were $f(0) = 0.0405$ and $f(1) = -0.1238$. Since the signs are opposites of each other, the Rachford–Rice function crosses zero between $\beta = 0$ and $\beta = 1$. Therefore, a physical two phase flash is possible at 90°C with the other operating conditions. At 95°C, the values are $f(0) = 0.2497$ and $f(1) = 0.0669$. Again, both signs are the same, meaning no root in the interval, so no two phase solution.

As shown by the results, temperature can have a significant effect on the phase condition. This goes to show how important it is to conduct a temperature study before attempting to calculate the vapor fraction: a difference of 5°C is the difference between a two phase solution existing and not.

4.4 Pressure Study

The effect of system pressure on the equilibrium behavior and phase separation of the mixture was studied while keeping temperature (90°C), feed flow rate (100 *kmol/hr*), and feed composition (0.40 ethanol) constant.



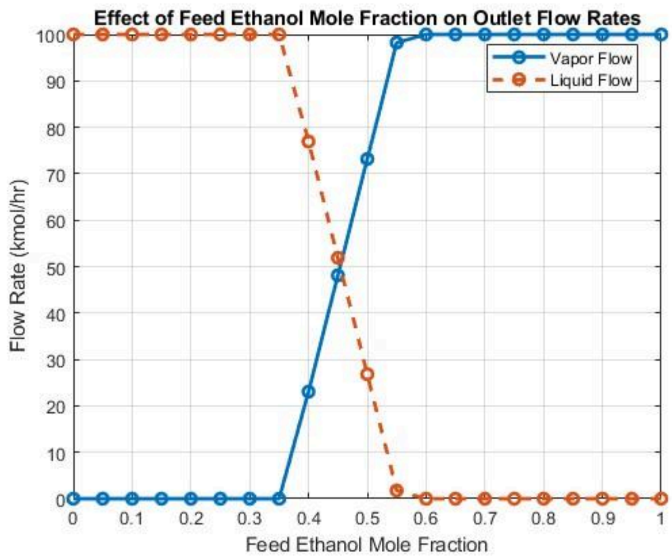
The effects of pressure change on the equilibrium ratio are studied through $K_i = \frac{P_i^{sat}}{P}$, where K is the ratio of the mole fraction of component i in the vapor phase to its mole fraction in the liquid phase, P_i^{sat} is the saturation vapor pressure of component i, and P is the total system pressure.

The saturation vapor of the components stay the same while the system pressure decreases and the temperature stays constant as well, meaning that increasing the system pressure results in a lower K-value. Additionally, a higher system pressure favors the liquid phase. This is because the vapor phase needs to reach a higher pressure for stability, and vice versa.

The behavior mentioned above is logical. The physical operation of flash separation reduces the pressure of the mixture rapidly, and the behavior shows that lowering the system pressure of a liquid mixture causes part of the liquid to vaporize as the equilibrium vapor pressure is higher than the surrounding pressure.

4.5 Feed Composition Study

The feed composition study was carried out to see how varying ethanol levels in the mixture would affect the vapor-liquid separation. The system pressure (760 mmHg), temperature (90°C), and feed flow rate (100 *kmol/hr*) were constant.



Increasing ethanol concentration is expected to have a higher amount of ethanol in the vapor phase when a two phase equilibrium exists, given that ethanol is more volatile than water. The flash calculation accounts for the feed composition through the Rachford–Rice equation:

$f(\beta) = \sum[\frac{z_i(K_i-1)}{1+\beta(K_i-1)}] = 0$, where z_i is the feed mole fraction of component i, hence the change of the Rachford-Rice function and the vapor fraction by changing the feed composition. The vapor composition is computed with $y_i = K_i x_i$.

4.6 Overall Discussion

The sensitivity studies conducted showed that temperature, pressure, and feed composition are responsible for changing the behavior of the system. Temperature was mainly responsible for the phase condition because increasing temperature increases the components' volatilities, increasing the pressure decreases the components' volatilities, and changing the feed composition alters the resulting phase compositions as well.

The results acquired from the studies align with the VLE relationships in the MATLAB model, which provides a useful representation of an idealized flash separation. However, it is entirely dependent on the assumptions made. To be more precise, ideal liquid phase is assumed for Raoult's law, and Antoine's equation is valid only within a certain temperature range. In reality, ethanol-water mixtures can show nonideal behavior, so it should be taken into account that the model is a simplification of industrial separation, not an exact prediction.

5.0 Conclusion and Future Work

This project successfully developed a MATLAB-based simulation to analyze vapor-liquid equilibrium and flash separation of an ethanol-water mixture. Multiple laws, equations, and assumptions were utilized to assist in simplifying the project, such as Raoult's law, Antoine's equation, K-values, bubble point, dew point calculations, and the Rachford-Rice equations to figure out phase behavior and compute vapor and liquid streams. To further simplify and organize the program, reusable MATLAB functions were built, like the flash separator and the Antoine's equation calculator.

The simulations observed how changes in operating conditions (temperature, pressure, feed composition, feed flow rate) affected the behavior of the system. Increasing temperature favored the vapor phase, and increasing pressure at a constant temperature favored the liquid phase.

The importance of combining thermodynamic relationship and numerical methods was an important lesson learned in this project. It is a helpful skill to get used to. The Rachford-Rice equation, for example, could only be executed within a certain boundary, so the checks prevented mathematically invalid vapor fractions as physical solutions. Additionally, working the program provided experience in translating engineering equations to MATLAB functions.

All in all, the project was very helpful as an educational tool. It allowed for operating conditions to be altered without manually repeating the calculations as functions were utilized to simplify everything. There are, however, limitations to the program. Ideal liquid phase behavior is assumed through Raoult's law, and there is a specific temperature range that cannot be exceeded to utilize Antoine's equation. The problem is that ethanol-water mixtures can show nonideal behavior, so the model is not a totally accurate simulation of a flash separator. Despite all that, the project helped build a foundation for the elaboration of the simulation to be more comprehensive.

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